FINAL PROJECT

IMPACT OF MENU DESIGN ON CUSTOMER ORDERING BEHAVIOR

MARKET STUDY

ETHNOGRAPHIC STUDIES IN
INTERACTION DESIGN
SEMESTER-4

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INTRODUCTION

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Menu design functions as a user interface within a physical service environment, shaping how customers interact with information.
It influences:

- Visual attention and scanning behavior
- Cognitive load and information processing
- Decision-making patterns

What is Menu Design?

Menu design refers to the structured presentation of information through:

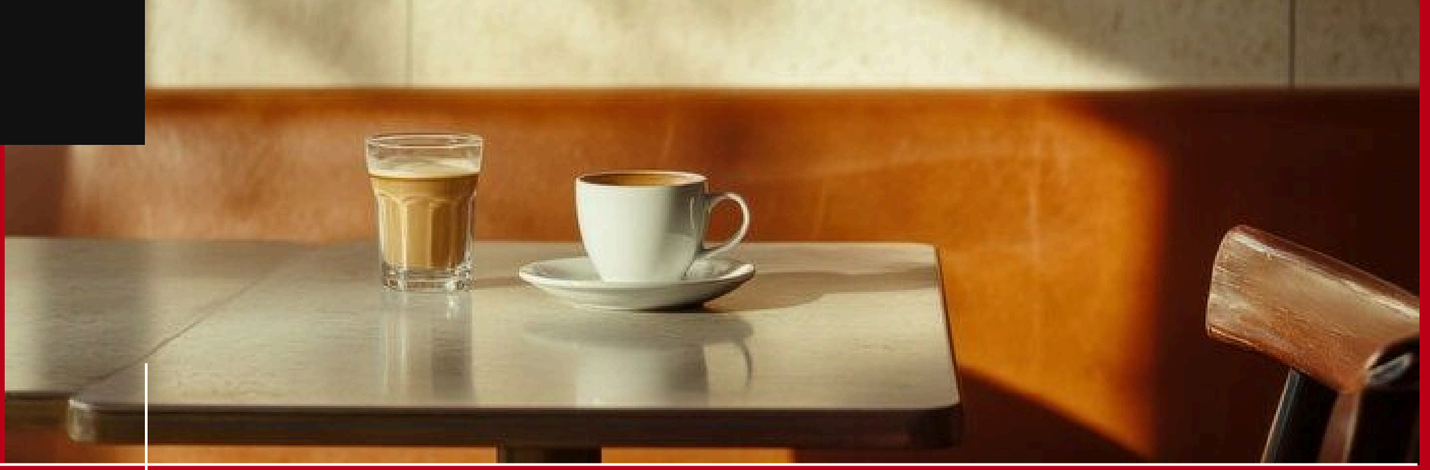
- Visual hierarchy
- Typography and readability
- Layout and spatial organization
- Use of imagery and color

It acts as a **decision architecture system guiding user choices.**

Why It Matters? (UX + Psychology)

Menu design impacts:

- **Perception** (what appears attractive or premium)
- **Cognition** (ease of understanding options)
- **Behavioral nudges** (highlighting, placement)
- **Decision efficiency** (time taken to choose)



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MARKET OVERVIEW

Overview of Markets visited

The study was conducted across three café market typologies:

- **Connaught Place** – premium, experience-driven cafés
- **Netaji Subhash Place** – mid-range, student-oriented cafés
- **Nearby local cafés** – budget-focused, high-frequency user

This segmentation enables **comparative behavioral analysis.**

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CATEGORISING MARKETS STUDIED

Market Type 1

Premium Cafés (Connaught Place)

Premium cafés are experience-driven spaces where menu design reinforces brand identity and aesthetics. Menus use strong visual hierarchy, curated layouts, and high-quality imagery to guide attention. The variety of options increases cognitive engagement, encouraging exploratory behavior. Behavioral nudges like highlights and specials influence choices, while pricing enhances perceived premium value. Here, the menu acts as both an informational and experiential design element.

Market Type 2

Mid-Range Cafés (Netaji Subhash Place)

Mid-range cafés balance usability and visual appeal, catering to students and young professionals. Menus are moderately structured with clear categorization, reducing decision fatigue and enabling quicker choices. Customers are more price-conscious and rely on familiarity. Selective use of visual cues (icons, highlights) supports efficient scanning. Overall, design focuses on clarity and goal-oriented decision-making.

Market Type 3

Local Cafés (Rohini)

Local cafés prioritize functionality and efficiency over aesthetics. Menus are simple, with limited options and low visual complexity, resulting in minimal cognitive load. Customers often rely on habitual choices, leading to faster decisions. There is a strong focus on price visibility, with little use of advanced design techniques. The menu serves primarily as an informational tool.

Menu design can create decision friction by increasing cognitive load and subtly influencing customer choices through visual hierarchy and design cues.

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Customers often experience decision friction while interacting with café menus due to poorly structured design. Issues such as cognitive overload, lack of clear visual hierarchy, and excessive or unorganized options increase the effort required to make a choice. Additionally, elements like images, highlights, and placement act as behavioral biases, subtly influencing decisions without conscious awareness. This makes menu design not just informational, but a key factor in shaping customer behavior and experience.

PROBLEM STATEMENT

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OBJECTIVES OF STUDY

The primary objective of this study is to analyze how menu design functions as a behavioral and interaction interface that influences customer ordering decisions within café environments. The study aims to examine the relationship between visual design elements such as layout, typography, imagery, color schemes, and pricing placement and how these factors shape user perception and attention. Additionally, the study seeks to understand how menu design impacts decision-making speed, cognitive load, and overall user experience, while comparing these effects across different café market segments to identify variations in user behavior, preferences, and interaction patterns.

The collected data was analyzed using both qualitative and quantitative approaches. Qualitative analysis included interpreting observations and interview responses to understand user behavior and experiences. Quantitative analysis involved identifying trends and patterns from survey data. Frameworks such as AEIOU (Activities, Environment, Interaction, Objects, Users) and empathy mapping were used to structure insights. Additionally, protocol analysis (think-aloud method) was applied to track user thought processes during decision-making.

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METHODS USED FOR ANALYSIS

Primary Research Methods

Primary data was collected directly from users through:

- Contextual Inquiry – observing real-time interaction with menus in café environments
- Ethnographic Observation – studying user behavior, social interactions, and environmental context
- Semi-Structured Interviews – gaining in-depth qualitative insights into user preferences and experiences
- Questionnaire (Survey) – collecting quantitative data on user behavior, choices, and perceptions

These methods helped capture first-hand user interaction, decision-making patterns, and behavioral responses.

Secondary Research Methods

Secondary data was gathered through analysis of existing materials and design artifacts:

- Artifact Analysis – evaluating menus as design objects based on visual hierarchy, typography, layout, color, and composition
- Study of existing menu design patterns and UX principles related to information hierarchy and user behavior

These methods helped in understanding design structures, patterns, and theoretical frameworks supporting the primary findings.

The integration of primary and secondary research ensured a holistic understanding of both user behavior and design effectiveness.

PROCEDURE OF ANALYSIS

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1. Selection & Field Visits

The research process began with selecting cafés across different market segments and conducting field visits to observe real-time user interactions with menus in natural settings.

2. Data Collection (Artifact Documentation)

During these visits, both printed and QR-based menus were documented and analyzed as design artifacts, focusing on layout, structure, and presentation styles.

3. Behavioral Observation

Observational data was recorded based on user behavior, decision-making time, navigation patterns, scrolling behavior, and hesitation points, providing insights into real interaction processes.

4. Data Organization

Survey and interview responses were compiled and categorized into key themes such as usability, perception, attention patterns, and decision triggers, enabling structured analysis.

5. Analytical Framework Application

The collected data was analyzed using comparative, behavioral, and perceptual frameworks, helping identify patterns and relationships between menu design elements and customer decision-making behavior.

Data was collected through field visits, observations, and surveys, then systematically analyzed using comparative, behavioral, and perceptual frameworks to identify patterns between design and user decisions.

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RESEARCH

Field Research

Field research involved direct observation of customers interacting with menus in cafés across Connaught Place and Netaji Subhash Place. Both printed menus and QR-based digital interfaces were studied to compare usability and engagement. Observations focused on decision-making time, scrolling and scanning behavior, hesitation points, and reliance on visual cues. This provided real-world insights into how users interact with menus in different contexts and environments.

General Observations

The study revealed that users rely heavily on visual hierarchy, imagery, and layout structure when navigating menus. Menus with excessive options or cluttered layouts tend to create cognitive overload, resulting in confusion and delayed decisions. Highlighted elements such as featured items, boxes, or images act as behavioral nudges, guiding user attention and influencing choices. Additionally, QR-based menus, although convenient, sometimes reduce usability due to navigation complexity, scrolling effort, and limited screen visibility.



PILOT STUDY

A pilot study was conducted prior to the main research to test the effectiveness and clarity of the research tools, particularly the questionnaire and interview approach. A small sample group of 8–10 participants (students and frequent café visitors) was selected to simulate the actual target audience.

Participants were asked to respond to the questionnaire and share feedback on question clarity, relevance, and ease of understanding. Minor interviews were also conducted to observe how comfortably users could interpret and answer the questions.

The pilot study revealed that some questions were ambiguous or repetitive, and certain terms required simplification. Based on this, the questionnaire was refined by improving wording, reducing redundancy, and ensuring a better flow of questions.

This step helped enhance the reliability and validity of the data collection process, ensuring that the final survey and interview methods were more effective and user-friendly for the main study.

Questions included:

The research included a set of structured and semi-structured questions aimed at understanding user behavior, perception, and decision triggers. Questions explored factors influencing ordering choices, the role of images and descriptions, ease of navigation, and points of confusion within menus. Users were also asked about their attention patterns, such as what they notice first and whether highlighted or visually emphasized items affect their final selection. These questions helped uncover both conscious and subconscious influences on decision-making.

These questions helped identify both conscious preferences and subconscious behavioral influences.

A structured set of questions was used to understand user behavior, perception, and decision-making triggers

QUESTIONS ASKED

QUESTIONS ASKED

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Questions for Customers (Users)

These focus on behavior, perception, and decision-making:

Ordering Behavior
- What factors influence your choice while ordering from a menu?
- How much time do you usually take to decide your order?
- Do you usually explore the menu or stick to familiar items?

Menu Interaction & Usability
- Do you find café menus easy to read and navigate? Why or why not?
- What do you notice first when you look at a menu (images, names, prices, layout)?
- Have you ever felt confused while ordering? What caused it?

Design Influence
- Do images or visuals in the menu affect your decision? How?
- Do highlighted or recommended items influence your choice?
- Do descriptions of items impact what you order?

Preferences
- Do you prefer simple menus or detailed ones?
- Do you prefer QR-based menus or printed menus? Why?
- Would you try a new item if it looks visually appealing on the menu?

Questions for Receptionist / Café Staff

These focus on business insights + observed customer behavior:

Customer Behavior Insights
- What are the most commonly ordered items on your menu?
- Do customers take time to decide or order quickly?
- Do customers often ask for recommendations?

Menu Effectiveness
- Are certain items highlighted or promoted on the menu? Why?
- Do you notice customers being influenced by images or special sections?
- Which type of menu (QR or printed) works better for customers here?

Operational Perspective
- Do customers ever seem confused while ordering? What do they usually ask?
- Have you made any changes to the menu based on customer feedback?
- Do highlighted or recommended items increase sales?

Design & Business Impact
- How important do you think menu design is for sales?
- Do you think simplifying the menu helps customers decide faster?
- Are premium items placed or presented differently?

ANALYSIS BASED ON RESPONSES

Analysis Based on Customer Responses

1. Decision-Making Behavior
Most customers rely on a mix of visual cues and familiarity while ordering. While some users explore the menu, a large portion prefers sticking to known items, especially when the menu is complex. Increased options lead to decision fatigue, causing delays or random selection.

2. Impact of Visual Elements
Images, highlights, and layout strongly influence choices. Items with images or placed in visually prominent sections receive more attention. This confirms that visual hierarchy and imagery act as primary decision triggers in menu interaction.

3. Usability & Navigation Issues
Many users reported difficulty navigating menus with dense layouts, unclear categorization, or small typography. QR-based menus introduced additional friction due to scrolling and screen limitations, indicating reduced usability in digital interfaces.

4. Attention Patterns
Customers tend to notice images, highlighted sections, and centrally placed items first. Price is often considered after initial attraction, showing that attention is driven visually before rational evaluation.

5. Preference Trends
Users showed a clear preference for simple, well-structured menus with limited options. While detailed menus offer variety, they often increase confusion, reinforcing the importance of clarity over complexity.

Analysis Based on Receptionist / Staff Responses

1. Ordering Patterns
Staff observations indicate that customers frequently order popular or highlighted items, suggesting that menu design directly influences sales trends. Repetitive ordering behavior also reflects habit formation and menu familiarity.

2. Decision Time & Behavior
Customers in premium cafés tend to take longer, exploring options, while in mid-range and local cafés, decisions are quicker. This highlights how menu complexity and context affect decision speed.

3. Role of Recommendations
Customers often ask for recommendations when menus are unclear, indicating gaps in menu communication and information hierarchy. Staff suggestions act as a substitute for weak design guidance.

4. Effectiveness of Menu Design
Staff confirmed that highlighted or promoted items are ordered more frequently, proving the effectiveness of behavioral nudges in menu design. Placement and visibility directly impact item selection.

5. Usability Challenges
Customers often ask questions about unclear items, combinations, or categories, indicating issues with menu clarity, labeling, and structure. This shows the need for better information organization.

Menu design plays a key role in shaping both customer perception and purchasing behavior, influencing not just what users choose but how they make decisions. The study shows that customers are guided by visual presentation as much as by personal preference.

Visual hierarchy, imagery, and highlighted elements were found to be the strongest influences on decision-making. Items that were visually prominent or emphasized attracted more attention and were chosen more frequently, indicating that design acts as a behavioral guide.

Menus with excessive options or cluttered layouts increased cognitive load, leading to confusion, longer decision times, and reliance on familiar items. In contrast, simple and well-structured menus improved clarity, navigation, and decision efficiency.

Staff responses further indicated that customers often seek help when menus are unclear, showing that staff recommendations compensate for weak menu design and highlighting the importance of effective information presentation.

Menu design acts as a silent decision-making tool, guiding attention, influencing choices, and impacting both user experience and sales.

COMBINED ANALYSIS (CUSTOMERS + STAFF)

OTHER ANALYSIS

A comprehensive visual, perceptual, and structural analysis of café menus was conducted to evaluate their effectiveness as communication tools. Typography was examined in terms of readability, font hierarchy, and legibility across viewing distances, while layout and spacing were analyzed to understand information hierarchy, grouping, and scanning flow. The use of color and imagery was studied for its role in capturing attention, shaping perception, and evoking emotional responses. Additionally, a comparative evaluation of digital (QR-based) and physical menus highlighted differences in usability, accessibility, interaction effort, and overall user engagement.

METHODS OF ANALYSIS

The collected data was analyzed using a combination of qualitative and quantitative research approaches. Qualitative analysis involved interpreting observational data and interview responses to understand user behavior, preferences, and pain points. Quantitative analysis focused on identifying patterns, trends, and frequency-based insights from survey responses. Analytical frameworks such as AEIOU (Activities, Environment, Interaction, Objects, Users) and empathy mapping were applied to structure user insights. In addition, protocol analysis (think-aloud method) was used to capture real-time cognitive processes and decision-making behavior during menu interaction.

RESULT & ANALYSIS

Results Acquired

The findings indicate that menu design has a direct and measurable impact on customer ordering behavior and decision efficiency. Visual elements such as images, highlighted sections, and strategic item placement significantly increased attention and item selection. Menus with clear structure and hierarchy improved navigation, comprehension, and speed of decision-making, while cluttered and dense layouts led to confusion, hesitation, and reliance on familiar choices. Overall, the results demonstrate that effective menu design enhances both user experience and business performance outcomes.

Comparative Analysis

The study revealed distinct behavioral patterns across different café markets. In premium cafés, customers exhibited exploratory decision-making, influenced by aesthetics, detailed descriptions, and presentation quality. In mid-range cafés, behavior was more efficient and goal-oriented, with users balancing visual appeal and practicality. In local cafés, decisions were quick, habitual, and familiarity-driven, due to simpler menus and limited options. These variations highlight how menu design responds to user context, expectations, and environmental factors.



The analysis highlights that menu design directly influences attention flow, perception, and decision-making efficiency. Visual hierarchy determines how users navigate and prioritize information, while imagery and highlighted elements act as strong attention and decision triggers. Excessive options increase cognitive load and decision fatigue, reducing efficiency, whereas structured and minimal layouts improve clarity, speed, and user confidence. Overall, menu design functions as a behavioral guide, subtly directing user choices.

KEY INSIGHTS

COMBINED FINAL ANALYSIS

Across all café markets, menu design functions as a behavioral and interaction system that shapes how users engage with choices. Premium cafés use design to enhance experience, perception, and exploration, mid-range cafés focus on clarity, balance, and efficiency, while local cafés prioritize speed, familiarity, and functional simplicity. Despite these contextual differences, a consistent pattern emerges: menu design significantly influences how users perceive options, process information, and make final decisions.

CONCLUSION OVERVIEW

The study establishes that menu design plays a crucial role in shaping customer behavior by influencing perception, cognition, and interaction patterns. A well-structured menu reduces decision friction, improves usability, and enhances the overall café experience, making it an essential component of service design and user experience strategy.

FINAL CONCLUSION

Menu design is not merely a visual element but a form of experience design and behavioral design that structures how users interact with choices. By guiding attention, simplifying complex information, and influencing perception, it directly impacts ordering behavior, user satisfaction, and business performance, positioning it as a critical tool in both design and hospitality contexts.

THANK YOU

